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Boosting CO_2 hydrogenation via size-dependent metal–support interactions in cobalt/ceria-based catalysts

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Metal–support interactions have a strong impact on the performance of heterogeneous catalysts. Specific sites at the metal–support interface can give rise to unusual high reactivity, and there is a growing interest in optimizing not only the properties of metal particles but also the metal–support interface. Here, we demonstrate how varying the particle size of the support (ceria–zirconia) can be used to tune the metal–support interactions, resulting in a substantially enhanced CO_2 hydrogenation rate. A combination of X-ray diffraction, X-ray absorption spectroscopy, near-ambient pressure X-ray photoelectron spectroscopy, transmission electron microscopy and infrared spectroscopy provides insight into the active sites at the interface between cobalt and ceria–zirconia involved in CO_2 hydrogenation to CH_4. Reverse oxygen spillover from the support during treatment in hydrogen results in the generation of oxygen vacancies. Stabilization of cobalt particles by ceria–zirconia particles of intermediate size leads to oxygen spillover to the support during the CO_2 and CO dissociation steps, followed by further hydrogenation of the resulting intermediates on cobalt.

The design of supported heterogeneous metal catalysts is of great importance for maximizing catalytic activity and controlling product selectivity[1]. Generally, the interaction between metal species and the support strongly influences the performance of catalysts. For reducible oxide supports (for example, TiO_2, Nb_2O_5 and CeO_2), these metal–support interactions (MSI) can have different origins[2]. For example, encapsulation of dispersed metal nanoparticles by a layer of oxide support material induced by reduction[3–8], oxidation[9,10] or adsorption of reactants[11] is referred to as a strong metal–support interaction (SMSI). Charge transfer from/to metal nanoparticles[12,13], support-induced restructuring of metal nanoparticles[14] and formation of specific metal–support interfaces[15] can also contribute to these MSI phenomena. Whereas SMSI often has a negative influence on the catalytic performance due to a decreased active metal surface area, the other effects can improve catalytic performance. The most well-known method to control the metal–support interactions is tuning the reduction temperature, resulting in a different extent of the decoration or encapsulation of metallic nanoparticles by a thin layer of reducible support[6,16]. Consecutive reduction–oxidation–reduction pretreatment can also be used to induce structural changes of metal particles supported on reducible supports such as TiO_2 and Nb_2O_5 (ref. [17]), leading to different adsorptive properties and catalytic performance. Li et al. demonstrated that NH_3 or H_2 treatment of anatase TiO_2 before impregnation with nickel can affect the decoration of metal particles in the final Ni/TiO_2 catalyst[18]. Zhang et al. reported the modification of a small Au-supported nanoparticle by Ti^{3+} precursors in aqueous solution, strongly influencing the stability of the catalyst[19].

Ceria (CeO_2) and ceria-based materials (such as ceria–zirconia (CZ), $CeZrO_4$) are widely used reducible oxide supports. Ceria-based materials differ from other supports because in addition to decoration or alloying, electronic phenomena also contribute to the observed MSI effects[20,21]. Lykhach et al. demonstrated that the dispersion of the active phase in Pt/CeO_2 systems can substantially affect charge transfer between the CeO_2 and Pt (ref. [22]). Pereira-Hernández showed that the support in Pt/CeO_2 catalysts, prepared by conventional impregnation and atom trapping methods, can be more easily reduced when the Pt–ceria interaction is stonger[23]. A model system, consisting of silver nanoparticles supported on ceria, exhibited unusual resistance against sintering due to the strong bonding of silver species on $CeO_2(111)$ terraces and oxygen defects[24]. The effect of the MSI in ceria-based catalysts is especially notable in CO_2 hydrogenation. Formation of Ce^{3+} sites in $Cu/CeO_x/TiO_2(110)$ and $CeO_x/Cu(111)$ facilitates CO_2 adsorption and was found to benefit methanol synthesis as compared to Cu(111) or $Cu/ZnO(000\bar{1})$ (refs. [25,26]). In CO_2 methanation by $Ni/CeZrO_4$ catalysts, it was shown that the pronounced redox behaviour of oxygen vacancies during transient experiments is linked to the formation of Ce^{3+} (ref. [27]).

Methanation of CO_2 is an important process used in the production of synthetic natural gas[28], purification of hydrogen feedstock for the production of ammonia[29] and reduction of CO_2 emissions[30]. Late transition metals such as Ni, Co and Ru supported on ceria-based materials are prominent methanation catalysts[31]. A number of studies described the influence of the metal–support interface on the performance of the Co/CeO_2 catalyst in CO_2 methanation[32], dry reforming of methane[33] and the (reverse) water–gas shift reaction[34,35]. An important corollary of these studies is that well-dispersed cobalt species on ceria, prepared by co-precipitation or sol–gel methods, exhibited an unusual reduction behaviour with several reduction stages in a broad temperature range. The occurrence of multiple reduction stages was attributed to the two types of metal species differing in the strength of their interaction with the support, to the surface and bulk reduction of Ce^{4+} to Ce^{3+} and to the reduction of the Co–Ce–O solid solution[34].

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In the present study, we present a simple method to control the metal–support interactions strength in Co/$CeZrO_4$ by varying the size of the support particles. We found that the preparation of cobalt nanoparticles on CZ supports with an intermediate particle size of ~20 nm leads to a higher dispersion of metallic cobalt as compared to CZ with larger and smaller particle sizes. We explain this finding by an optimal strength of the cobalt–CZ interaction, resulting in enhanced stability of the cobalt nanoparticles during the reduction step. The advantages of these catalysts with an optimized cobalt–support interaction include a high extent of ceria reduction at low temperature and a drastically improved catalytic activity in the C–O bond dissociation. For catalysts with an optimal MSI, we demonstrate that facile formation of oxygen vacancies in ceria and oxygen spillover are pivotal for the high CO_2 methanation activity.

Results

Dependence of activity on the support particle size. The particle size of the CZ support was varied in a range of 9–120 nm by increasing the calcination temperature of the initial precipitate between 500 °C and 1,000 °C (Supplementary Fig. 1 and Supplementary Table 1). The obtained CZ materials were used to prepare CoCZ (10 wt% cobalt) catalysts (Supplementary Fig. 2) by wet impregnation with hexaamminecobalt(III) complex, $(Co(NH_3)_6)^{3+}$, in a strong electrostatic adsorption mode[36]. This preparation method led to catalysts with a similar initial particle size of Co_3O_4 upon calcination (Fig. 1a,b and Supplementary Fig. 3). This cannot be achieved with other, more common low-melting Co precursor salts because of calcination-induced agglomeration[37]. Typically, optimal catalytic activity is achieved by using supports with a large surface area (small particles) that allow a high dispersion of the active metal[38]. However, we observed that cobalt nanoparticles supported on CZ with the smallest particle size (CZ500) exhibit a relatively poor catalytic performance. The methanation activity of CoCZ catalysts displays a maximum at the intermediate size of the CZ support particles (Fig. 1c). The highest conversion and methane yield were observed for the CoCZ700 catalyst. The CO_2 conversion at 225 °C with CoCZ700 was 5.8 and 2.9 times higher than with CoCZ500 and CoCZ1000, respectively. The same trend was observed at higher reaction temperatures (Supplementary Figs. 4 and 5).

The unusual activity trends can be due to a change in the cobalt particle size but also to the formation of different crystal phases of the CZ support[39,40]. CZ exists in monoclinic, cubic and tetragonal forms depending on the cerium concentration and the calcination temperature[41]. For the prepared CZ materials, we observed a shift from the monoclinic to the cubic and tetragonal phases at calcination temperatures above 700 °C (Supplementary Fig. 6). To determine the influence of the crystal phase on the catalytic performance, we compared cobalt catalysts supported on CZ700 with cobalt supported on a commercial CZ support. The surface areas of these support materials as determined by N_2 physisorption are similar, while the commercial support contains exclusively tetragonal CZ (Supplementary Fig. 7). The catalysts prepared from these two supports show similarly high catalytic activity with a CO_2 conversion of approximately 50% at 300 °C (Supplementary Fig. 8). This result clearly demonstrates that there is no influence of the support's crystal phase on the catalytic performance. Change of activity can also be related to the partial decoration of cobalt nanoparticles by oxidic support after high-temperature reductive pretreatment[42]. We ruled out this possibility by performing the reduction at lower temperature, which had no influence on the catalytic performance of the catalyst (Supplementary Fig. 9). We also compared these data to the catalytic activity of cobalt supported on TiO_2 with a particle size equal to that of Co_3O_4 (Supplementary Figs. 10 and 11). Co/TiO_2 catalysts are known to be active catalysts for CO_2 methanation[7,43,44]. Although the activity of CoCZ500 was comparable to that of the Co/TiO_2 catalyst, the CoCZ700 catalyst performed much better in CO_2 methanation. The activity of Co- and Ni-based catalysts reported in the literature is shown in Supplementary Fig. 12 and Supplementary Table 2. To summarize, the CoCZ700 catalyst shows enhanced catalytic activity in CO_2 methanation as compared to cobalt supported on CZ supports with smaller and larger particle sizes and to another common methanation catalyst, Co/TiO_2. Moreover, CoCZ700 demonstrates higher activity at low temperature (200–275 °C) than typical Ni-based catalysts.

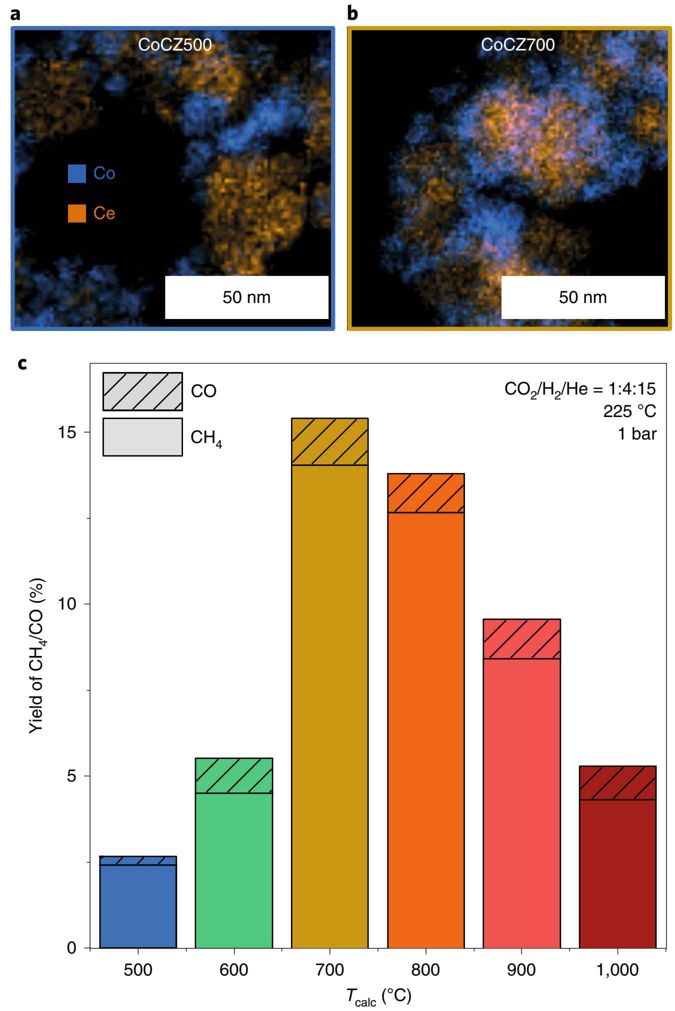

Fig. 1 | The effect of support calcination pretreatment on catalytic activity. **a**,**b**, EDX mapping of as-prepared CoCZ500 (**a**) and CoCZ700 (**b**). **c**, Catalytic performance at 225 °C of cobalt-based catalysts supported on CZ calcined at different temperatures (T_{calc}).

Catalytic activity is linked to the extent of MSI. MSIs can be an alternative explanation for the volcano-like activity trend of CoCZ catalysts with respect to the CZ particle size. To elucidate the MSI effect, we first analysed the reducibility of CoCZ catalysts by temperature-programmed reduction (TPR) analysis. The CoCZ500 sample exhibits two typical reduction features corresponding to the stepwise reduction of Co_3O_4 nanoparticles, that is $Co_3O_4 \xrightarrow{\beta} CoO \xrightarrow{\delta} Co$. TPR profiles of CoCZ700 and CoCZ800 contained additional features (Fig. 2a). The low-temperature feature (α) might be attributed to the partial surface reduction of the ceria[33]. Complete reduction of these samples is shifted to a substantially higher temperature, which is characteristic for cobalt catalysts with SMSI[45].

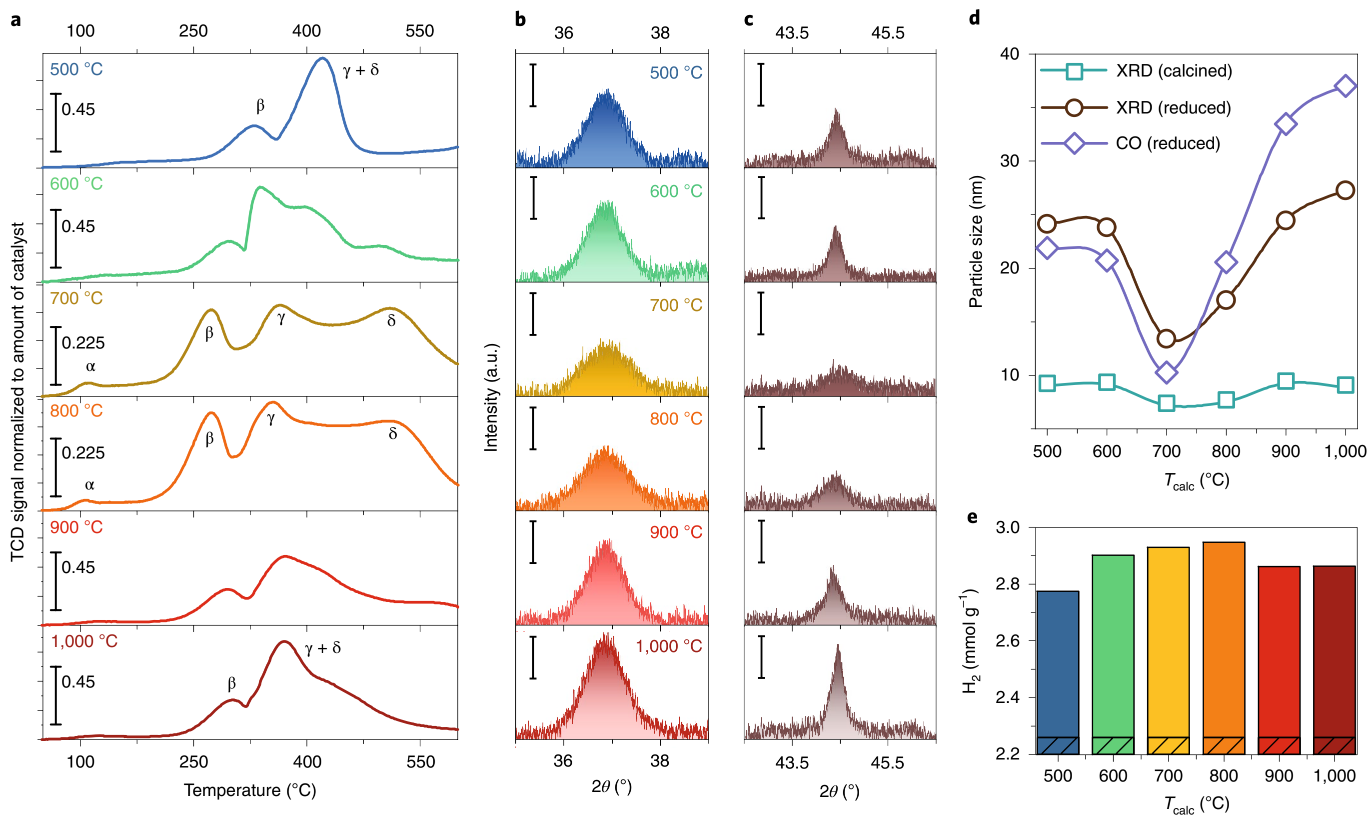

Fig. 2 | Characterization of CoCZ catalysts. a, TPR profiles of CoCZ calcined at 500–1,000 °C (TCD signal is normalized to the amount of catalyst). **b**,**c**, X-ray diffraction of CoCZ calcined at 500–1,000 °C (**b**) and reduced at 500 °C (**c**). Black scale bars, 200 counts. **d**, Cobalt particle size estimated by X-ray diffraction for calcined and reduced samples and by CO chemisorption for reduced samples. **e**, Quantification of hydrogen consumption during TPR experiments (solid bar, excess of H_2 reduction; hatched bar, amount required for complete Co_3O_4 reduction).

Moreover, the high-temperature reduction region is split into two features, which can be assigned to the reduction of cobalt species weakly (γ) and strongly (δ) interacting with the support[42]. Other explanations of the extra high-temperature features can be the reduction of the cobalt–support interface Co–O–Ce species[46] or hydrogen spillover resulting in bulk reduction of the support[33]. A pronounced excess of H_2 consumption (Fig. 2e) observed for the catalysts with enhanced MSI points to a deeper reduction of the support ($Ce^{4+} \rightarrow Ce^{3+}$). The reduction behaviour of cobalt species weakly and strongly interacting with the support was followed by in situ X-ray absorption near-edge spectroscopy (XANES) analysis. Supplementary Figs. 13 and 14 illustrate the shift of Co_3O_4 reduction to a lower temperature for the CoCZ700 sample, in line with the TPR profiles.

Stabilization of the intermediate CoO phase in a wider temperature range suggests that the γ feature in the TPR profile is related to reduction of the support rather than to CoO reduction. Indeed, the $CoO \rightarrow Co^0$ transformation occurred at higher temperature for CoCZ700, which is typical for systems with a strong MSI[33]. TPR and XANES data link the catalytic activity of the samples and the extent of the MSI—the most active catalysts display complex reduction behaviour with high-temperature features related to both cobalt and cerium oxides.

Strong interfacial anchoring of Co nanoparticles on CZ support. An important aspect of CoCZ catalysts is the stabilization of cobalt nanoparticles by the support under high-temperature reducing conditions. X-ray diffraction data of the CoCZ catalysts show that the dispersion of the initial cobalt oxide is similar with particle sizes in the 7–9 nm range (Fig. 2b,d). This finding is in line with scanning transmission electron microscopy–energy-dispersive X-ray spectroscopy (STEM-EDX) results (Fig. 1a,b and Supplementary Fig. 3). After reduction in H_2 at 500 °C, however, we observed substantial differences in the cobalt metal particle size (Fig. 2c). As derived from X-ray diffraction and CO chemisorption measurements, the size of the reduced cobalt particles increased substantially (Fig. 2d).

Transmission electron microscopy (TEM) analysis shows that before reduction, cobalt is well dispersed in its oxide form, independent of the particle size of the support (Fig. 3a,c). Upon reduction, the cobalt phase sinters more substantially for the CoCZ500 sample (Fig. 3b,d), in line with CO chemisorption and X-ray diffraction results. The formation of notably larger cobalt particles (>50 nm) was also observed by STEM-EDX (Fig. 3f and Supplementary Fig. 15). Weak MSI in CoCZ500, manifested by the absence of high-temperature TPR features, are a likely reason for the sintering. Large cobalt agglomerates were observed for the CoCZ900 sample as well, which is due to the low surface area of the support (Fig. 3h and Supplementary Fig. 15). The CoCZ700 catalyst demonstrated a much better sintering resistance during high-temperature reduction. Nearly identical particle size distributions were obtained for as-prepared and reduced CoCZ700 samples, and no larger agglomerates were observed (Fig. 3c,d,g and Supplementary Fig. 15). Besides thermal stability, the CoCZ700 sample exhibits improved catalytic stability in CO_2 methanation at 300 °C (Supplementary Fig. 16). The presented data show that the optimum strategy to prepare efficient CoCZ catalysts is to use supports with large enough particle sizes to facilitate the strong interfacial MSI without sacrificing surface area (Fig. 3e).

Interface chemistry of CoCZ catalysts. Strong MSI in ceria-based materials are often linked to the formation of surface oxygen

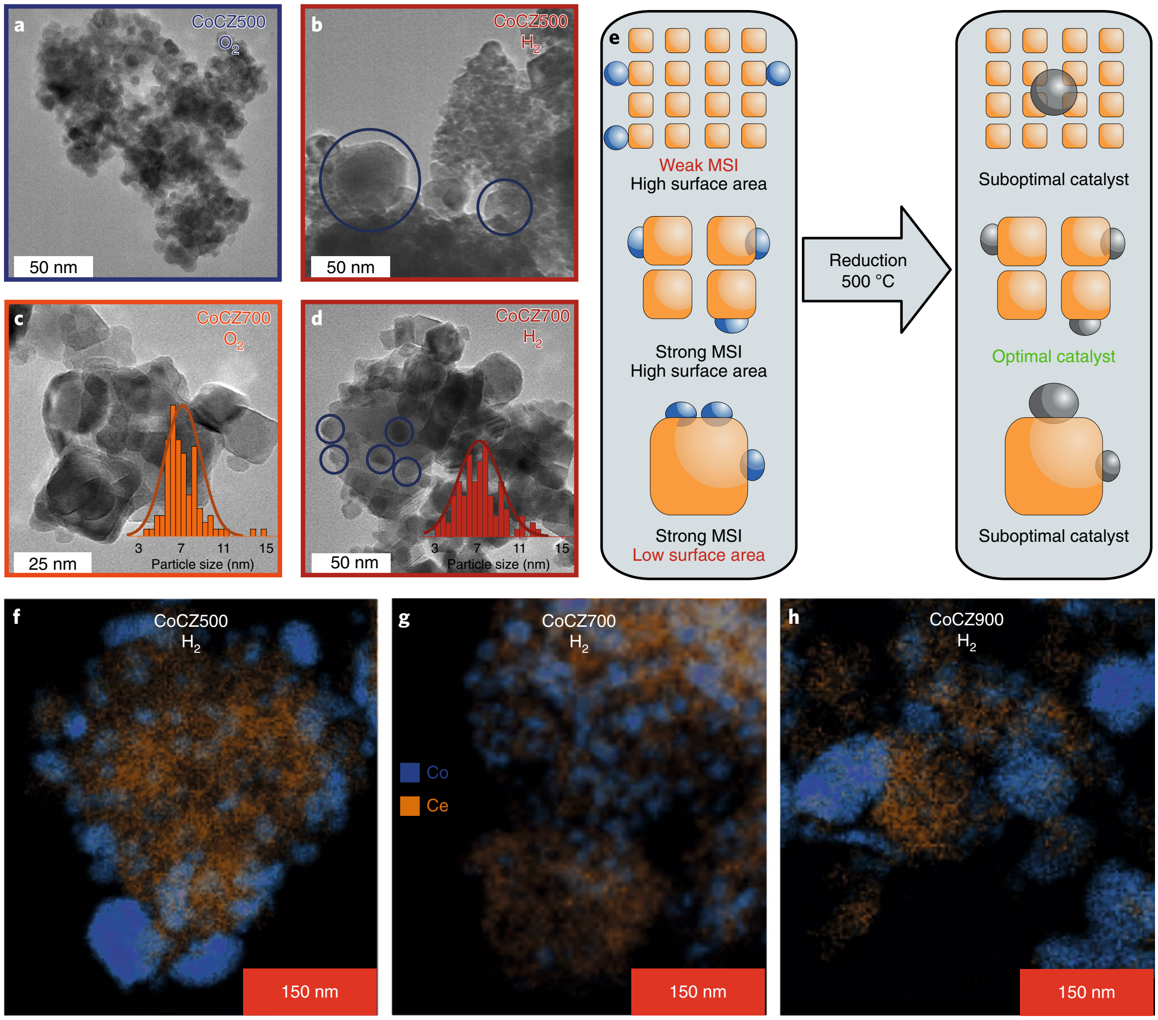

Fig. 3 | The relationship between catalyst morphology and cobalt sintering. a–d, TEM images of calcined (**a**,**c**) and reduced (**b**,**d**) cobalt samples prepared on CZ calcined at different temperatures; insets show particle size distribution of CoCZ700 after calcination and after reduction at 500 °C. Blue circles represent Co nanoparticles. **e**, Schematic representation of the interplay between the MSI and support particle size on the properties of CoCZ catalysts. Blue and grey spheres represent oxidized and reduced cobalt particles, respectively; orange particles represent the CZ support. **f–h**, STEM-EDX images of samples reduced at 500 °C: CoCZ500 (**f**), CoCZ700 (**g**) and CoCZ900 (**h**).

vacancies. Although TPR is commonly used to probe the reducibility of ceria, this technique lacks the chemical specificity to follow the formation of oxygen vacancies. Oxygen vacancies in ceria can be directly detected by Raman[47], infrared[27,48] and resonant photoemission spectroscopies[49]. Here we studied the formation of vacancies using infrared spectroscopy. For this purpose, we focused on the broad band in the 2,115–2,135 cm^{-1} region, corresponding to the $^2F_{5/2} \rightarrow {}^2F_{7/2}$ electronic transition of Ce^{3+} (ref. [48]). Enhanced MSI facilitate the formation of surface vacancies starting at a temperature of about 150 °C (Fig. 4b–d). It correlates with the intensive α-feature in the TPR profiles of CoCZ700 and CoCZ800. The onset of vacancy formation for CoCZ500 is shifted to a higher temperature in the 200–250 °C range, which explains the absence of the low-temperature reduction feature in the corresponding TPR profile. Reduction of the bare supports occurs at a much higher temperature, in line with the TPR results (Supplementary Fig. 17). Clearly, the Co–CZ MSI influences not only the cobalt but also the support. In the literature, the removal of active oxygen at low temperature is assigned to the reduction of the metal–support interface[50]. Facile formation of oxygen vacancies can be beneficial for the catalytic activity of systems where the support is actively involved in the catalytic cycle. As the CZ support is known to play an important role in adsorption and activation of CO_2 on the metal–support interface[27], dynamic redox properties of oxygen vacancies are essential for the catalytic performance. Transient experiments at 300 °C followed by operando infrared spectroscopy demonstrate the fast formation/re-oxidation of oxygen vacancies for CoCZ700 in H_2 and the reaction mixture, while these changes are negligible for the bare support (Supplementary Fig. 18). These results confirm the relevance of the facile formation of oxygen vacancies to the catalytic performance of CoCZ systems in CO_2 hydrogenation.

The reduction behaviour of both cobalt and ceria was followed by near-ambient pressure X-ray photoelectron spectroscopy (NAP-XPS) analysis of the most relevant CoCZ700 sample (Supplementary Fig. 19). As shown in Fig. 4f, a substantial fraction of Ce^{4+} is reduced to Ce^{3+} before metallic cobalt can be detected. A high reduction degree of ceria was observed at 450 °C, simultaneously with the formation of a small fraction of Co^0 required for hydrogen spillover onto the support. Based on these data and XANES analysis, we can attribute the pronounced γ-feature in the TPR profile of CoCZ700 to the reduction of the support. Since cobalt remains oxidized during the reduction of the support at low

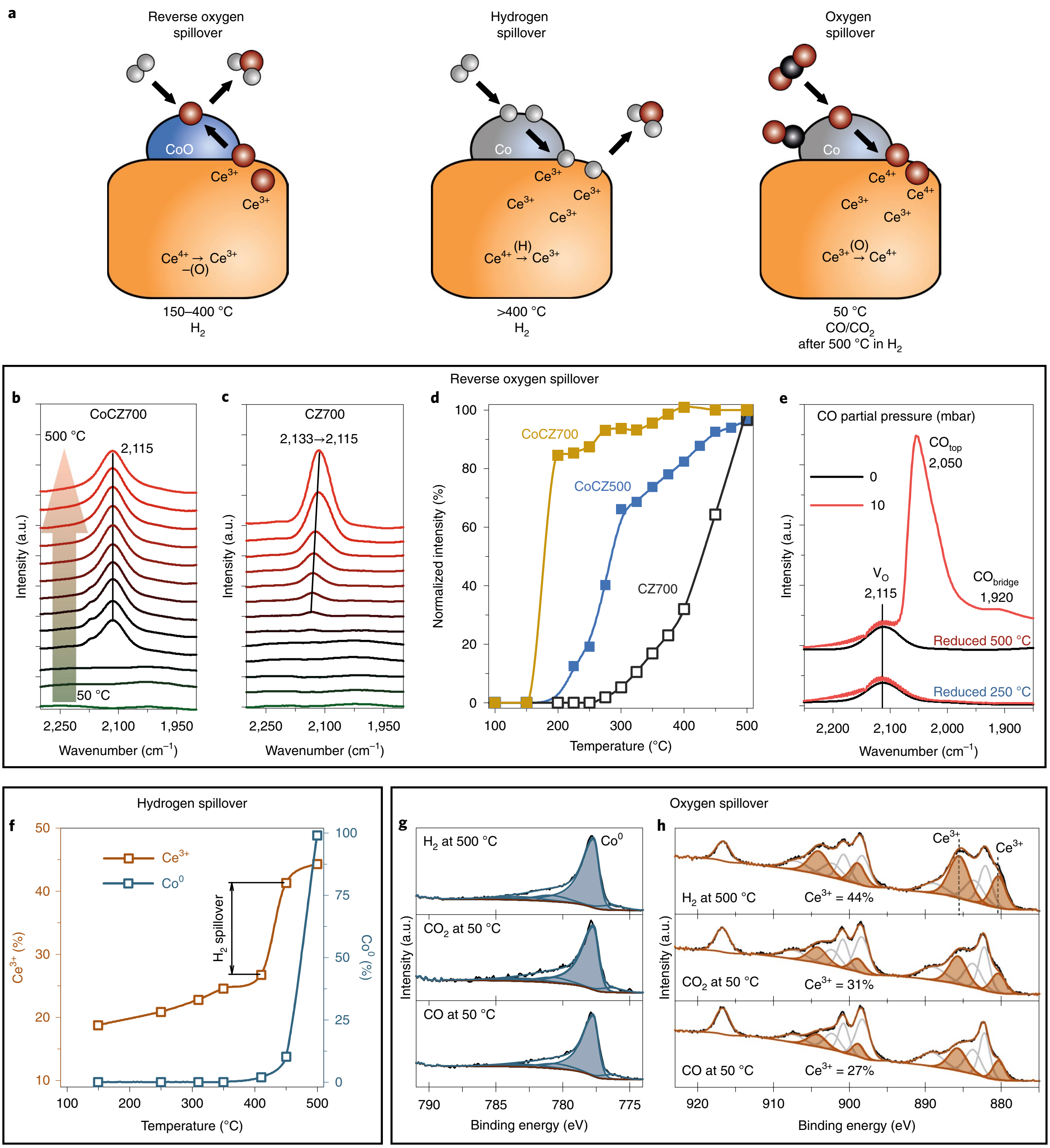

Fig. 4 | Spillover effects observed by means of operando NAP-XPS and infrared spectroscopy. **a**, Schematic representation of the spillover effects in the CoCZ700 catalyst. Light grey, red and black spheres represent H, O and C atoms, respectively. **b**,**c**, Oxygen vacancy formation evaluated by intensity of the band at 2,115 cm^{-1}; spectra obtained for CoCZ700 (**b**) and support (**c**). **d**, Temperature dependence of normalized intensity of oxygen vacancy band for CoCZ700, CoCZ500 and CZ700. **e**, CO adsorption on CoCZ700 reduced at 250 °C and 500 °C; spectra obtained in vacuo and after exposure to 10 mbar of CO; the bands located at 1,920 cm^{-1} (CO_{bridge}), 2,050 cm^{-1} (CO_{top}) and 2,115 cm^{-1} (V_O) are assigned to the bridge and on-top CO adsorption sites on metallic cobalt and oxygen vacancy, respectively. Black curve, in vacuo; red curve, 10 mbar CO. **f**,**g**,**h**, TPR of CoCZ700 followed by NAP-XPS. Spectra obtained for reduced CoCZ700 at 500 °C and after exposure to CO and CO_2 atmosphere at 50 °C: Co 2*p* (**g**) and Ce 3*d* (**h**).

temperature, this finding points to reverse oxygen spillover from the support to cobalt oxide[51,52]. The more conventional hydrogen spillover mechanism of support reduction[53] can be excluded in this temperature region on the basis of infrared spectroscopy of adsorbed CO (Fig. 4e). CoCZ700 reduced at 250 °C contained oxygen vacancies (2,115 cm^{-1} band), while no metallic cobalt sites were observed.

After high-temperature reduction, on the other hand, the intensity of the band related to oxygen vacancies remained unchanged and a broad CO adsorption band located at 2,050 cm^{-1} was observed, indicating the presence of metallic cobalt.

Exposure of the reduced catalyst to CO or CO_2 at 50 °C caused notable oxidation of the support, while cobalt remained metallic without any noticeable changes (Fig. 4g,h). The observation of carbonyl groups on the metallic cobalt surface by infrared spectroscopy proves that the dissociation of the C–O bond upon CO_2 exposure at 50 °C takes place (Supplementary Fig. 20). Exposure of CoCZ700 to CO_2 during transient experiments at 300 °C results in substantial oxidation of surface Ce^{3+} species (Supplementary Fig. 18). These findings confirm that oxygen spillover to the support takes place upon the dissociation of CO and CO_2 molecules on the optimized cobalt surface and at the Co–CZ interface of the CoCZ700 catalyst. For catalysts with suboptimal MSI, CO_2 dissociation is less pronounced than for CoCZ700, as followed from in situ infrared measurements (Supplementary Fig. 21). The catalytic activity correlates to the CO_2 dissociation ability rather than to the cobalt dispersion (Supplementary Fig. 22). Therefore, the intrinsic activity of the Co–CZ interface for the dissociation of C–O bonds is a relevant descriptor for the performance of CoCZ catalysts in CO_2 hydrogenation.

Conclusions

We show a simple method to control the extent of MSIs in cobalt/CZ systems. Cobalt nanoparticles supported on small particles of CZ are prone to sintering during the high-temperature reduction required for activation of the catalyst. Counter-intuitively, cobalt nanoparticles are much stronger when stabilized on CZ supports with an intermediate particle size in the 20–30 nm range. Enhanced metal–support interactions benefit cobalt dispersion after high-temperature reduction, resulting in the highest CO_2 methanation activity. Detailed characterization by XANES, NAP-XPS and infrared spectroscopy demonstrated dynamic properties of the Co–CZ interface. Directly observed phenomena include facile surface oxygen vacancy formation via reverse oxygen spillover, reduction of ceria via hydrogen spillover and healing of oxygen vacancies upon CO/CO_2 exposure via oxygen spillover. The rich chemistry of the optimized Co–CZ interface enables the stabilizing of metal nanoparticles under high-temperature reducing conditions and enhanced CO_2 activation, and can find potential applications in steam and dry methane reforming, preferential oxidation, the (reverse) water–gas shift, the Fischer–Tropsch process and other related catalytic reactions.

Methods

All chemicals were obtained from Sigma-Aldrich: cobalt(II) acetate tetrahydrate, ammonia solution (30% NH_3 in H_2O), cerium(III) nitrate hexahydrate and zirconium(IV) oxychloride octahydrate.

Preparation of CZ supports. CZ supports were prepared by co-precipitation. An aqueous solution of metal precursors was prepared from $ZrOCl_2$•$8H_2O$ and $Ce(NO_3)_3$•$6H_2O$. Afterwards, it was precipitated by adding an aqueous solution of ammonia (30%) dropwise at room temperature. The pH of the solution was adjusted to 9. The resulting precipitate was aged for 24 h, followed by filtration, washing and drying in a convection oven at 110 °C overnight. To prepare CZ supports with different particle sizes, dried precipitate was calcined in an air flow at temperatures between 500 °C and 1,000 °C for 4 h. The obtained samples are denoted as CZ500 to CZ1000, corresponding to the calcination temperature.

Preparation of CoCZ catalysts. A series of CZ-supported cobalt catalysts with different supports was prepared using a wet impregnation method. For this purpose, the desired amount of cobalt acetate was dissolved in 30 mL of ammonia solution (30 wt%). The suspension obtained by adding 2 g of CZ to the solution was stirred for 2 h, after which water was removed by evaporation. The catalysts were dried in air at 110 °C overnight and then calcined at 350 °C for 4 h. Prior to the catalytic activity measurements, the catalysts were reduced at 500 °C in H_2 for 4 h.

Characterization. The crystalline structure of the catalysts was determined by recording **X-ray diffraction** patterns with a Bruker D2 Phaser diffractometer using Cu Kα radiation. The particle size was estimated with the Scherrer equation. Reduced samples were transferred to the sample holder via a glovebox. The samples were placed in an X-ray diffraction sample holder and covered with Kapton tape.

The metal content was determined using inductively coupled plasma optical emission spectrometry (ICP-OES) with a Spectro CIROS charge coupled device (CCD) spectrometer. Prior to measurement, the samples were dissolved in a 1:2.75 (by weight) mixture of $(NH_4)_2SO_4$/(concentrated) H_2SO_4 and diluted with water.

Hydrogen TPR (H_2-TPR) experiments were performed with a Micromeritics Autochem II 2920 instrument. Typically, 100 mg of catalyst was loaded in a tubular quartz reactor. The sample was reduced in 4 vol.% H_2 in N_2 at a flow rate of 50 mL min^{-1}, while heating from room temperature up to 600 °C at a rate of 10 °C min^{-1}.

N_2 physisorption was performed at −196 °C on a Micromeritics TriStar II 3020 to determine the specific surface area (SSA) of the CZ supports. The samples were degassed at 160 °C for at least 12 h prior to N_2 physisorption measurements. Surface areas were calculated using the Brunauer–Emmett–Teller (BET) method.

TEM images were acquired with a Tecnai 20 transmission electron microscope (FEI) equipped with a LaB6 filament and operated at an acceleration voltage of 200 kV. Calcined catalysts were prepared by dropping a suspension of finely ground material in analytical grade absolute ethanol onto Quantifoil R 1.2/1.3 holey carbon films supported on a copper grid. Reduced catalysts were transferred to an argon-filled glovebox and dispersed in dry *n*-hexane; then a few droplets were placed on Cu TEM grids. The grid was transported in a GATAN vacuum transfer holder (model number CHVT3007).

The morphology and the nanoscale distribution of elements in the samples was studied using STEM-EDX. Measurements were carried out on an FEI cubed Cs-corrected Titan operating at 300 kV. Calcined samples were crushed, sonicated in ethanol and dispersed on a holey Cu support grid. Reduced catalysts were passivated before sample preparation. Elemental analysis was done with an Oxford Instruments EDX detector X-Max^N 100TLE.

CO chemisorption measurements were carried out using a Micromeritics ASAP 2010C instrument. Before each measurement, the samples were dried in vacuum at 110 °C. Samples were subsequently heated in flowing H_2 with a rate of 10 °C min^{-1} to the final reduction temperature of 500 °C. A reduction time of 4 h was used, after which the samples were evacuated at 520 °C for 60 min. The CO adsorption isotherms were measured at 30 °C. The CO/Co ratios at zero pressure were determined by extrapolation of the linear part of the isotherm. Particle size estimations are based on the assumption of hemispherical geometry, assuming a CO/$Co_{surface}$ adsorption stoichiometry of 1.5.

Operando NAP-XPS measurements were carried out on a SPECS system. All spectra were obtained using monochromatized Al Kα irradiation (1,486.6 eV) generated by an Al anode (SPECS XR-50) and an excitation source power fixed at 50 W. XPS measurements at pressures up to ~20 mbar are possible owing to a differential pumping system, which separates the electron analyser (SPECS Phoibos NAP-150) from the reaction area. The aperture of the nozzle is 0.3 mm. The gas supply to the reaction cell consisted of calibrated mass-flow controllers, providing a total flow of 2.5 mL min^{-1} (standard pressure and temperature). The reaction pressure was kept at 2 mbar by a back pressure controller. For reduction, the gas mixture contained 0.5 mL min^{-1} of Ar and 2 mL min^{-1} of H_2. All the gases used were of high purity (99.999%). A standard residual gas analyser (QMS) located in the differential pumping stage of the analyser allowed us to follow the gas phase composition while recording the photoelectron spectra. The total acquisition time of the survey spectrum, including the O 1*s*, C 1*s*, Ce 3*d* and Co 2*p* regions, was around 60–70 min. A pass energy of 40 eV was typically used with a step size of 0.1 eV and a dwell time of 0.5 s. The binding energies of the Co 2*p* and Ce 3*d* regions were corrected to the U‴ (Ce^{4+}) component of the Ce 3*d* line with a characteristic position of 916.7 eV[54–56]. The position and shape of this peak is independent of the atomic ratio of Ce^{3+} to Ce^{4+} (as long as Ce^{4+} can be detected), allowing reliable energy calibration of the photoelectron spectra at different reaction conditions. To estimate the atomic ratios, a standard procedure, including use of atomic sensitivity factors and subtraction of the Shirley background, was applied. Spectral lines were fitted using a symmetric pseudo-Voigt function referred to as GL(30) using the CasaXPS software. The Ce 3*d* line was fitted according to a model described elsewhere[55,57]. All the spectra are presented as recorded, meaning that no normalization or other manipulations were performed.

Infrared spectra were recorded on a Bruker Vertex 70v Fourier transform infrared spectrometer equipped with a deuterated-triglycine sulfate (DTGS) detector. The experiments were performed in situ by using a home-built environmental transmission infrared cell. Self-supporting pellets were made by pressing approximately 12 mg of a sample in a disc with a diameter of 13 mm. Each spectrum was collected by averaging 64 scans with a resolution of 2 cm^{-1} in the 4,000–1,000 cm^{-1} range. Oxygen vacancy formation was studied according to the following procedure: (1) the samples were calcined before measurement at 400 °C in order to remove most of carbonate species; (2) after calcination, the samples were cooled down to 50 °C and exposed to a H_2/N_2 flow; and (3) then, the samples were reduced at different temperatures in the 50–500 °C range with a ramp rate of 10 °C min^{-1} to follow the intensity of the Ce^{3+} characteristic band at 2,115 cm^{-1}. All spectra were recorded at 50 °C to avoid background changes and temperature-induced features.

XANES was used to study the oxidation state of the Co phase during catalyst reduction. Measurements were done at the Co K edge (7.7 keV) in transmission mode at beamline B18 of the Diamond Light Source. The energy was selected with a Si(111) monochromator. Energy calibration was done using a Co foil ($E_0 = 7.709$ keV). The photon flux of the incoming and outgoing X-ray beam was determined using two ionization chambers. The obtained X-ray absorption spectra were background-subtracted and normalized using Athena software. Linear combination fitting of operando data was performed using three independently recorded reference spectra of a Co foil and CoO and Co_3O_4 powders. In a typical experiment, ~15 mg of the catalyst sample diluted with boron nitride was placed in a tubular quartz reactor, as described elsewhere[58]. Catalysts were reduced in this operando cell by heating at a rate of 10 °C min^{-1} to 550 °C followed by an isothermal dwell of 20 min in a flow of 10 vol.% H_2 in He at a total flow rate of 50 mL min^{-1}.

Catalytic activity. Catalytic activity measurements were performed in a ten-tube parallel microflow reactor. The samples were pressed, sieved and crushed, and the fraction between 125 μm and 250 μm was used. Each quartz reactor was filled with 50 mg of sample diluted with 200 mg of SiC of the same sieve fraction. The obtained mixture was enclosed between two quartz wool plugs. The reaction was performed at atmospheric pressure. Prior to reaction, the catalysts were reduced in situ in a flow consisting of 10 vol.% H_2 in He (total flow rate 50 mL min^{-1} at STP per reactor tube), while heating from room temperature to 500 °C at a rate of 10 °C min^{-1}, followed by an isothermal dwell of 4 h. After cooling to the reaction temperature in the same flow, the pretreatment gas was replaced by a feed consisting of 5 vol.% CO_2 and 20 vol.% H_2 in He (total flow rate 50 mL min^{-1} at STP). The temperature was increased by steps of 25 °C at a rate of 5 °C min^{-1}. When the target temperature was reached, a period of 20 min was allowed for stabilization. Then, the effluent gas was analysed by online gas chromatography with an Interscience Compact gas chromatograph equipped with PLOT (thermal conductivity detector (TCD)) and MolSieve (TCD) columns.

Data availability

The data that support the findings of this study are available from the authors upon reasonable request.

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Author contributions
A.P. synthesized and characterized the set of CZ samples (TPR, X-ray diffraction, and CO chemisorption and infrared spectroscopy). E.H.O. performed the catalytic measurements. V.M. and A.P. performed the operando NAP-XPS experiments and interpreted the results. N.K., V.M. and A.P. performed and interpreted the operando X-ray absorption spectroscopy measurements. A.J.F.H. performed the TEM measurements with an in situ holder. T.F.K. synthesized and provided the cobalt–titania sample. A.P., N.K. and E.J.M.H. wrote the paper. All authors discussed the results and commented on the manuscript.

Competing interests
The authors declare no competing interests.

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SUPPLEMENTARY INFORMATION

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Boosting CO_2 hydrogenation via size-dependent metal–support interactions in cobalt/ceria-based catalysts

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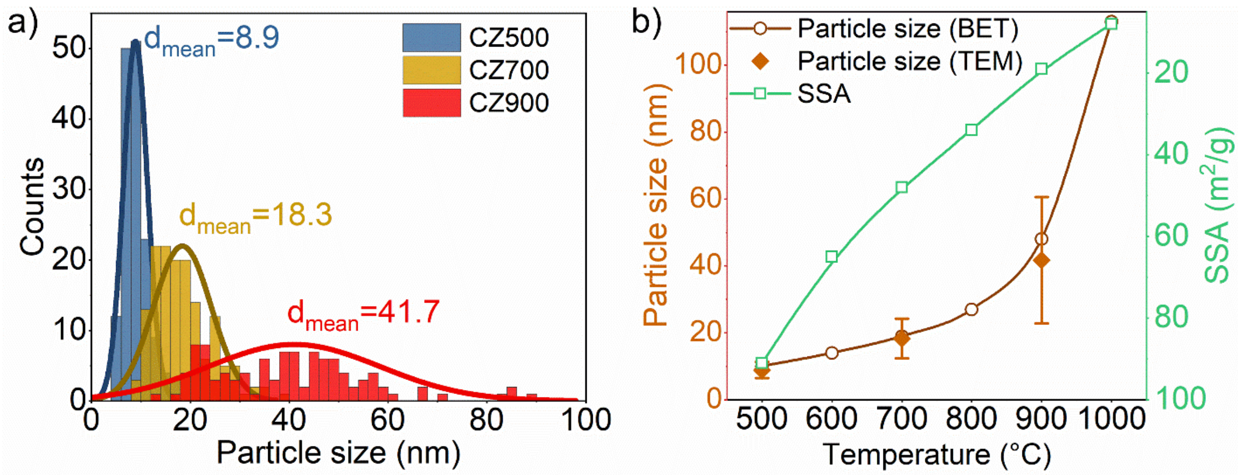
Supplementary information

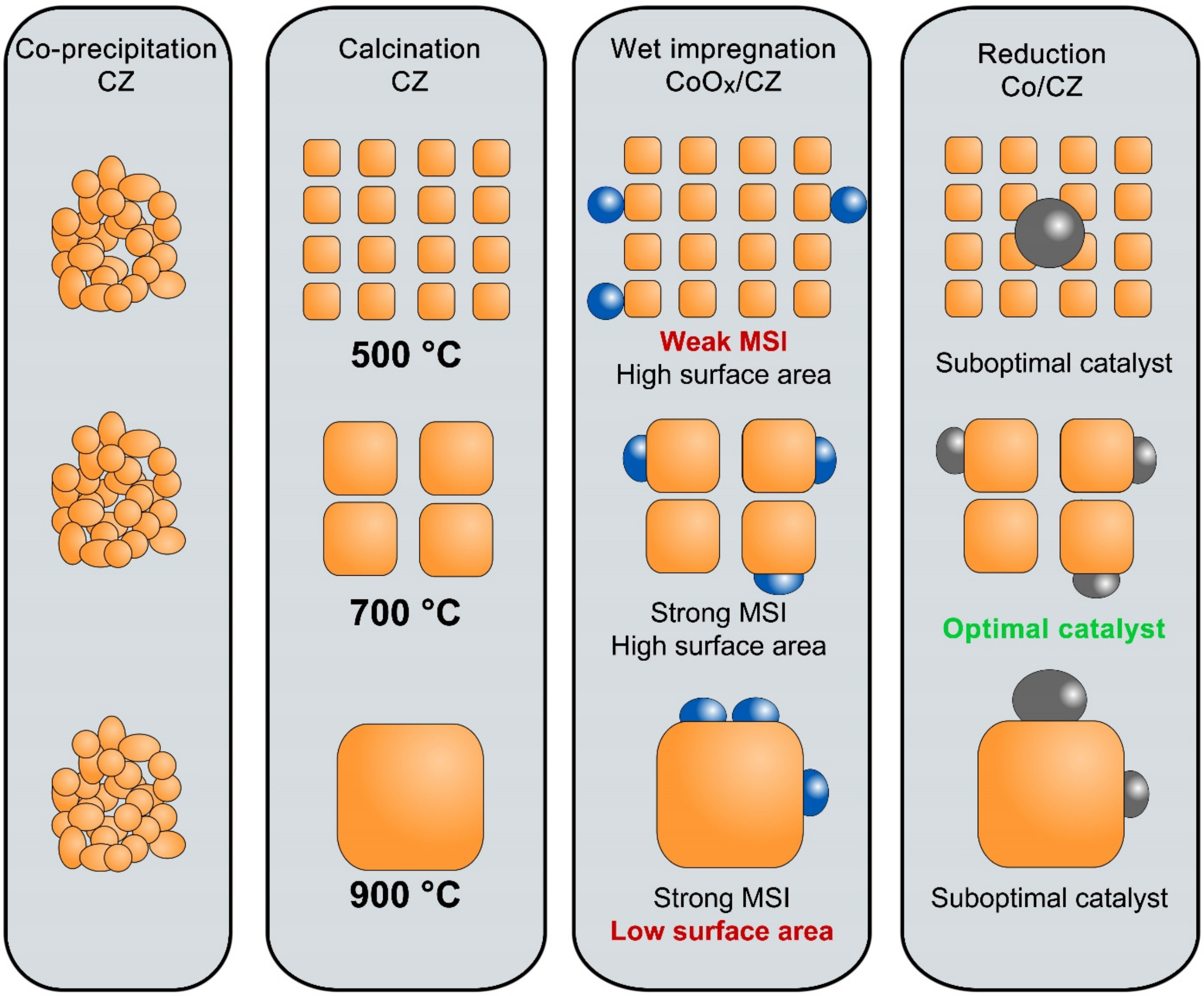


Supplementary Figure 1. a) Particle size distribution and mean particle size obtained from TEM images for CZ500, CZ700 and CZ900; b) Comparison of support particle size obtained by BET and TEM. Error bars represent the standard deviation in support mean particle sizes determined by TEM.

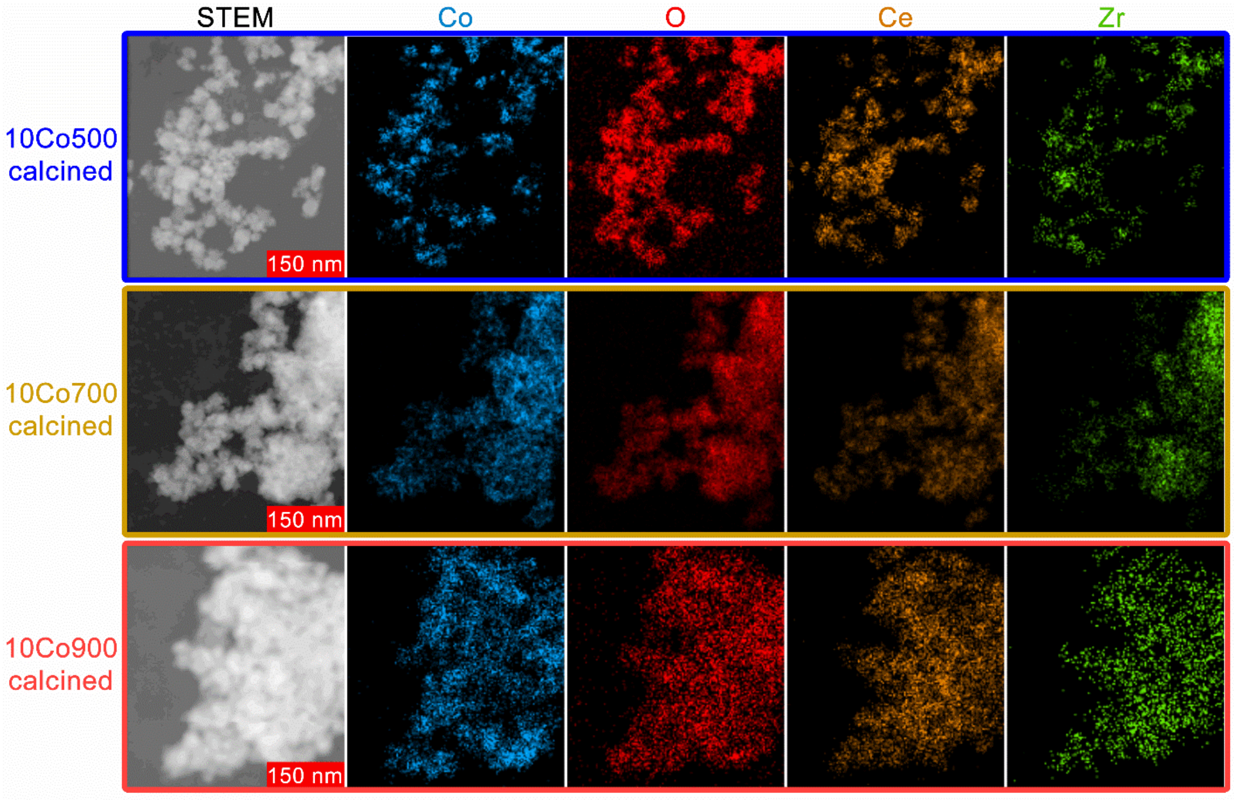
Supplementary Table 1. BET surface area and particle size of the supports.

Support	BET, m^2/g	Particle size[a], nm
CZ500	91	10
CZ600	65	14
CZ700	48	19
CZ800	34	27
CZ900	19	48
CZ1000	8	113
CZ SA	54	17
TiO_2 P25	58	25

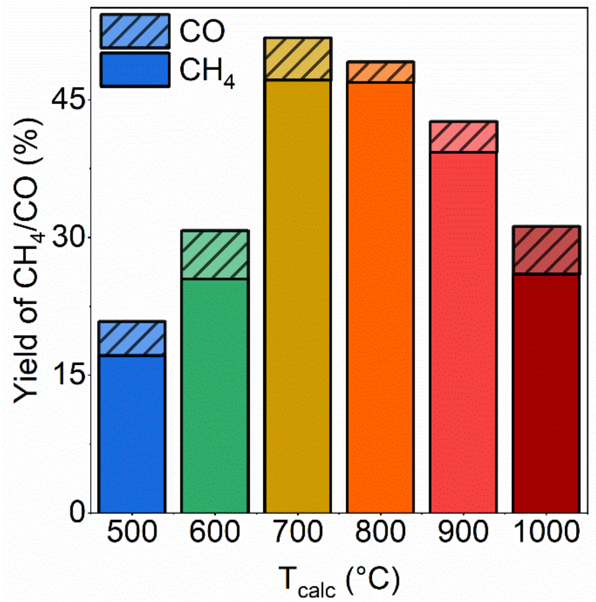
[a] estimated from BET surface area with assumption that particles are non-porous and spherical. Density of ceria-zirconia – 6.61 g/cm^3. Density of TiO_2 – 4.23 g/cm^3.



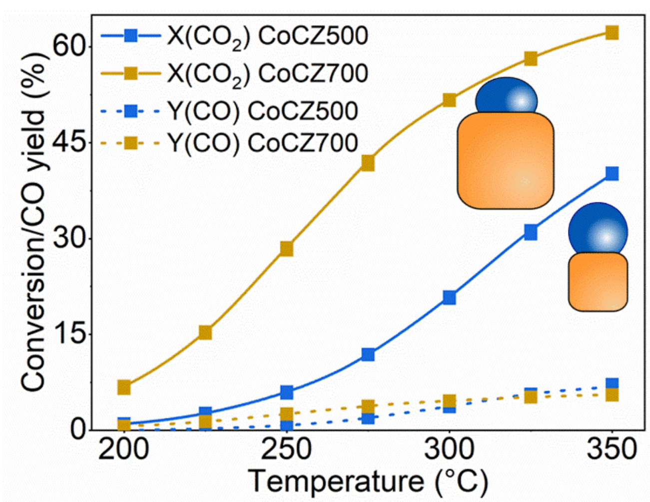
Supplementary Figure 2. Catalysts preparation procedure involving following steps: co-precipitation of ceria-zirconia, calcination of obtained material at different temperatures, impregnation with cobalt precursor and calcination at 350°C, reduction at 500°C.



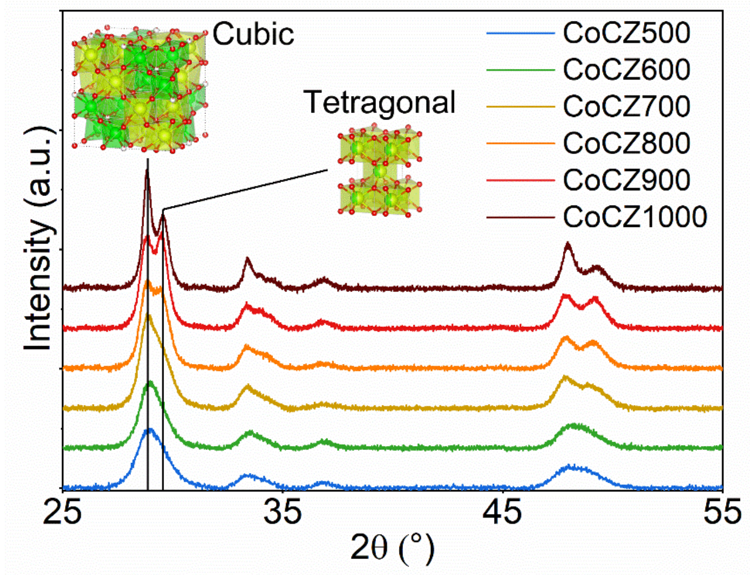
Supplementary Figure 3. STEM images and EDX mapping (Co, O, Ce, Zr) of calcined cobalt catalyst: CoCZ500, CoCZ700 and CoCZ900.



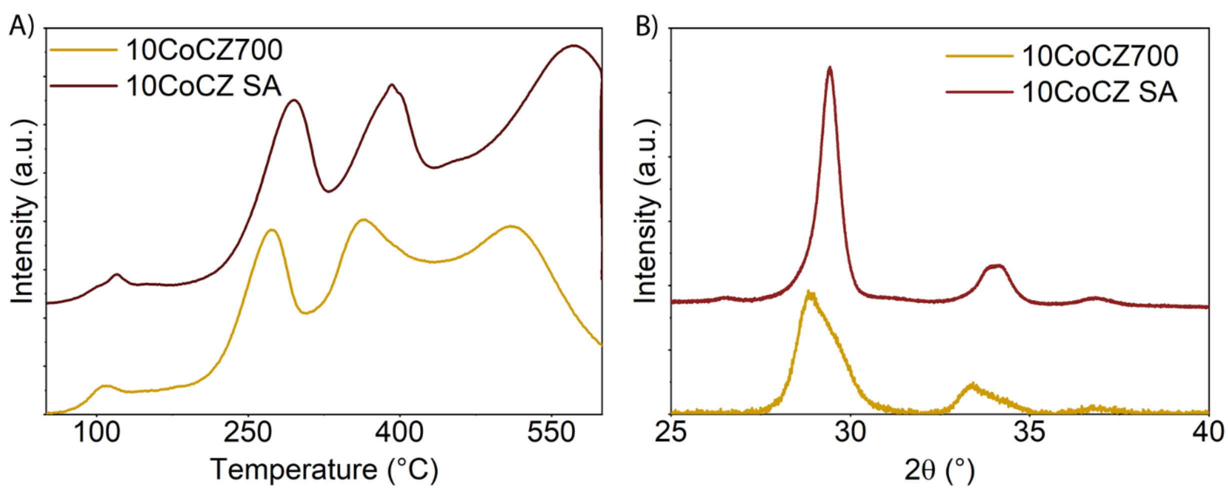
Supplementary Figure 4. Catalytic performance at 300 °C of cobalt based catalysts supported on CZ as a function of CZ calcination temperature.



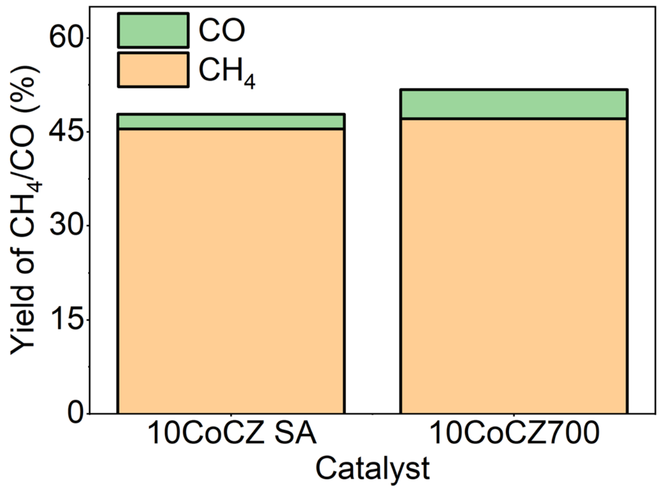
Supplementary Figure 5. CO_2 conversion and CO yield as a function of reaction temperature for CoCZ500 and CoCZ700.



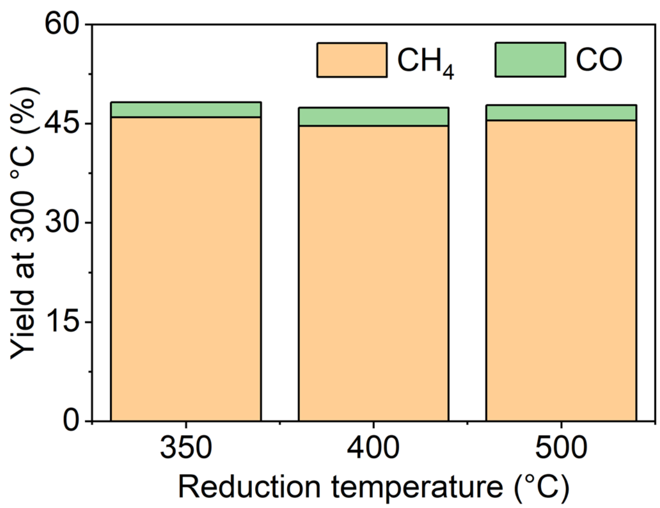
Supplementary Figure 6. XRD patterns of calcined CZ samples.



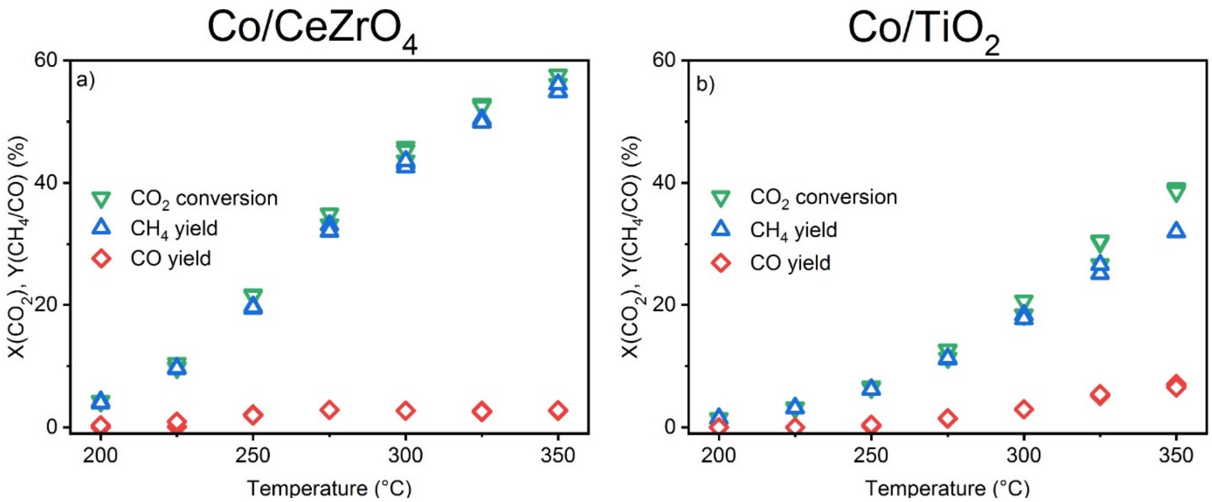
Supplementary Figure 7. Comparison of CoCZ700 and CoCZ SA: A) TPR profiles; B) XRD patterns.



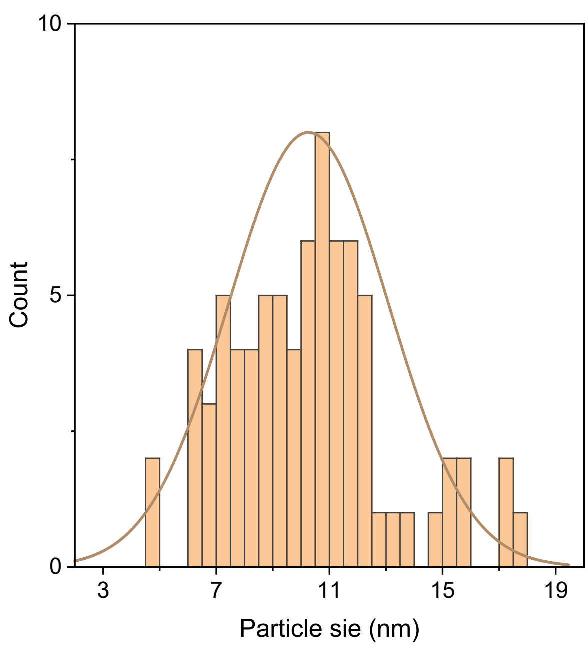
Supplementary Figure 8. Influence of crystal phase on catalytic activity: catalytic activity at 300 °C of CoCZ700 and CoCZ SA.



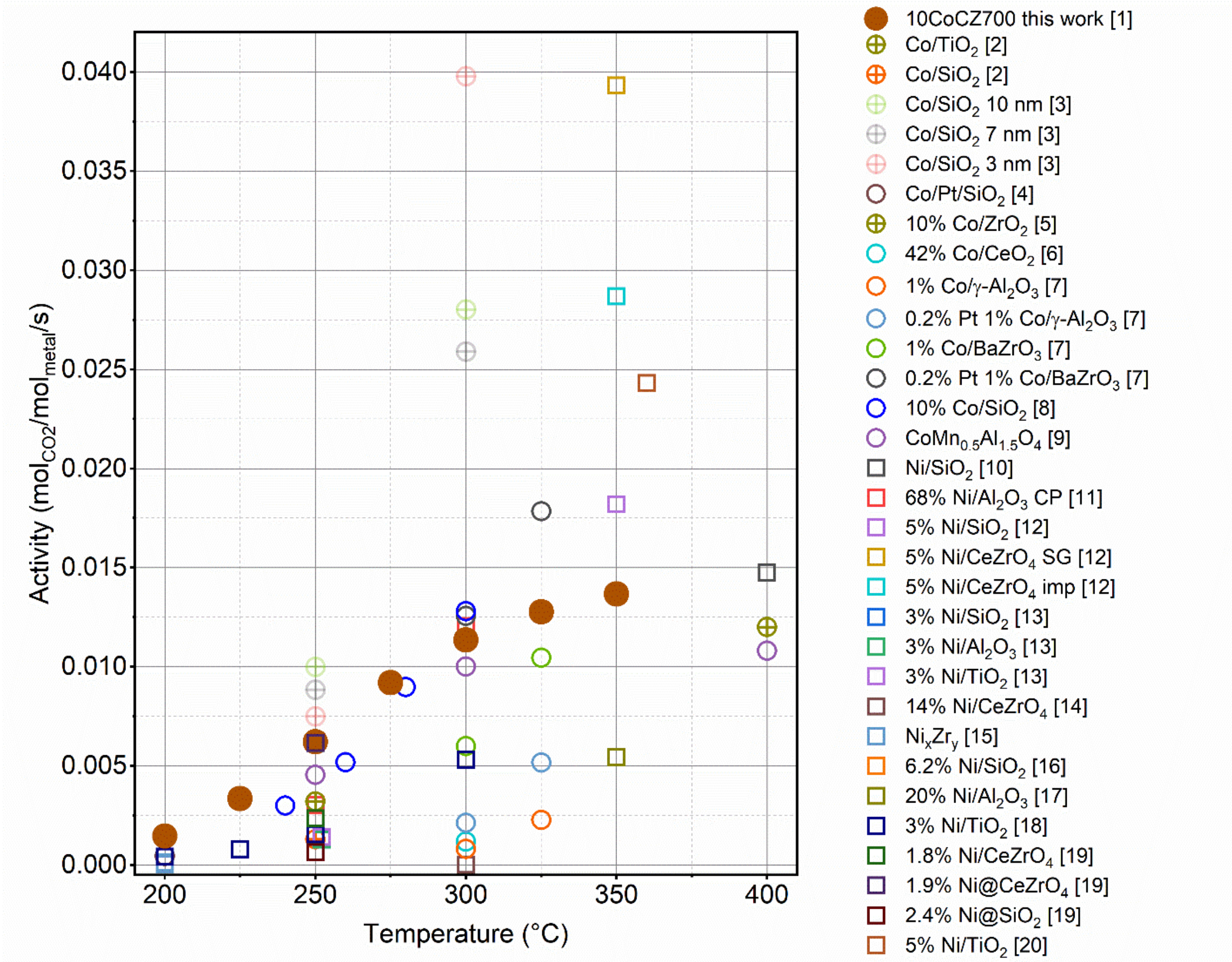
Supplementary Figure 9. Yield of CO and CH_4 in CO_2 methanation over 10CoCZ-SA catalysts reduced at different temperature. Conditions: 1 bar, 300 °C, 50 mg of catalyst, 5 % CO_2, 20 % H_2, He balance, total flow of 50 mL/min.



Supplementary Figure 10. Catalytic performance of 10Co/CZ (commercial) and 8.25Co/TiO_2 (P25).



Supplementary Figure 11. Particle size distribution of cobalt for calcined 8.25Co/TiO_2.



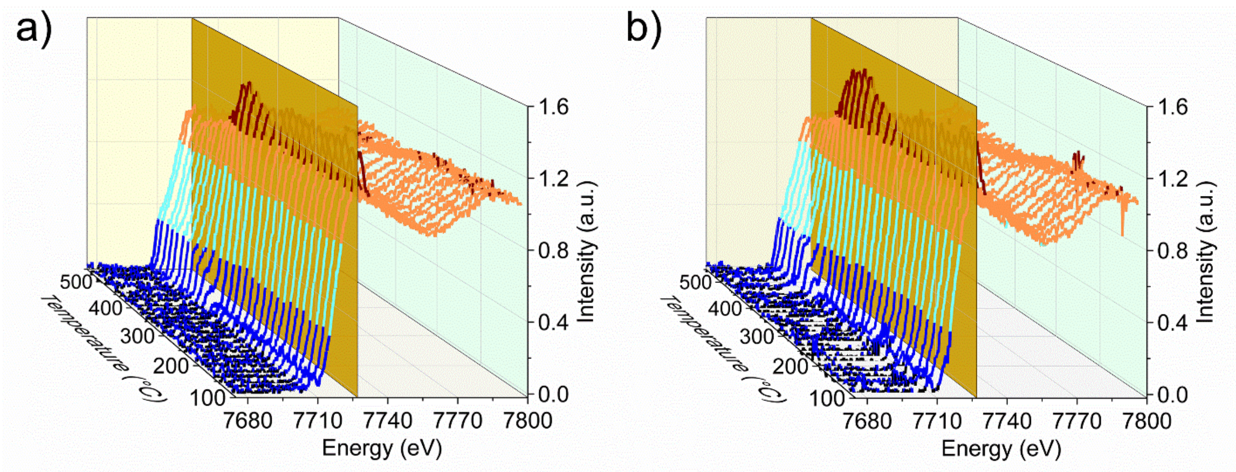
Supplementary Figure 12. Activity of catalysts in CO_2 methanation normalized to the total amount of metal at different temperatures. ○ – cobalt-based catalysts; ⊕ – cobalt-based catalysts at elevated pressure; □ – nickel-based catalysts. Numbers in the legend correspond to the position of catalysts in the table below.

Supplementary Table 2. Activity of nickel and cobalt-based catalysts in CO_2 methanation.

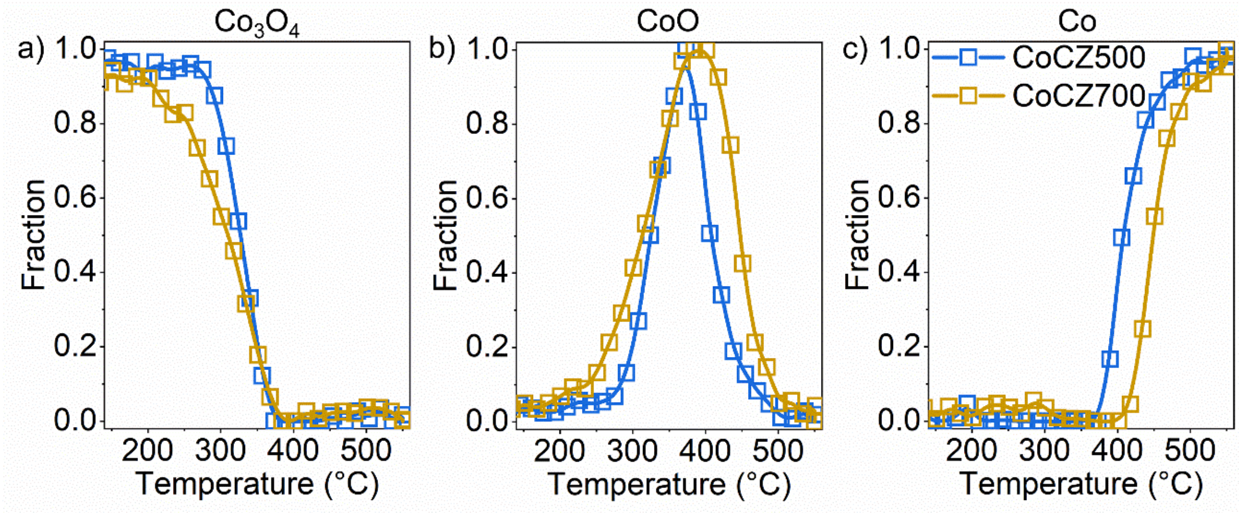
	Catalyst	CO_2 conc., %	Conditions Remarks	T_{reac}, °C	Specific activity, $mol_{CO2}/mol_{metal}/s$	Ref.
1	10CoCZ700	5	1 bar	200 225 250 275 300 325	1.46E-03 3.35E-03 6.22E-03 9.19E-03 1.13E-02 1.28E-02	This work

				350	1.37E-02	
2	Co/TiO_2	25	Oxidized, 5 bar	250	3.20E-03	1
	Co/SiO_2		Reduced, 5 bar	250	1.30E-03	
3	Co/SiO_2 10nm	22.2	6 bar	250 300	1.00E-01 2.80E-01	2
	Co/SiO_2 7 nm			250 300	6.20E-02 1.81E-01	
	Co/SiO_2 3nm			250 300	2.25E-02 1.19E-01	
4	$Co/Pt/SiO_2$	22.2	1 bar	200	4.99E-04	3
5	10% Co/ZrO_2	20	30 bar	400	1.20E-02	4
6	42% Co/CeO_2	10	1 bar	300	1.18E-03	5
7	1%Co/γ-Al_2O_3	3	1 bar	300 325	8.13E-04 2.27E-03	6
	0.2%Pt1%Co/γ-Al_2O_3			300 325	2.12E-03 5.17E-03	
	1%Co/$BaZrO_3$			300 325	5.99E-03 1.05E-02	
	0.2%Pt 1%Co/$BaZrO_3$			300 325	1.26E-02 1.78E-02	
8	10%Co/SiO_2	10	1 bar	240 260 280 300	3.00E-03 5.18E-03 8.98E-03 1.28E-02	7
9	$CoMn_{0.5}Al_{1.5}O_4$	20	1 bar	250 300 400	4.55E-03 1.00E-02 1.08E-02	8
10	Ni/SiO_2	10	1 bar	400	1.48E-02	9
11	68%Ni/Al_2O_3 CP	19	2 bar	250 300	3.01E-03 1.21E-02	10
12	5%Ni/SiO_2 5%Ni/$CeZrO_4$ SG 5%Ni/$CeZrO_4$ imp	16.4	1 bar	350 350 350	1.82E-02 3.93E-02 2.87E-02	11
13	3%Ni/SiO_2 3%Ni/Al_2O_3 3%Ni/TiO_2	1	1 bar	252 252 252	1.40E-03 1.29E-03 1.43E-03	12
14	14%Ni/$CeZrO_4$	20	1 bar	300	1.48E-05	13
15	$Ni_{70}Zr_{30}$ $Ni_{60}Zr_{40}$ $Ni_{50}Zr_{50}$ $Ni_{40}Zr_{60}$ $Ni_{30}Zr_{70}$ 4%Ni/ZrO_2	20	1 bar	200 200 200 200 200 200	3.46E-05 7.19E-05 1.02E-04 7.06E-05 4.50E-05 1.24E-04	14
16	6.2%Ni/SiO_2	20	1 bar	450	5.32E-03	15

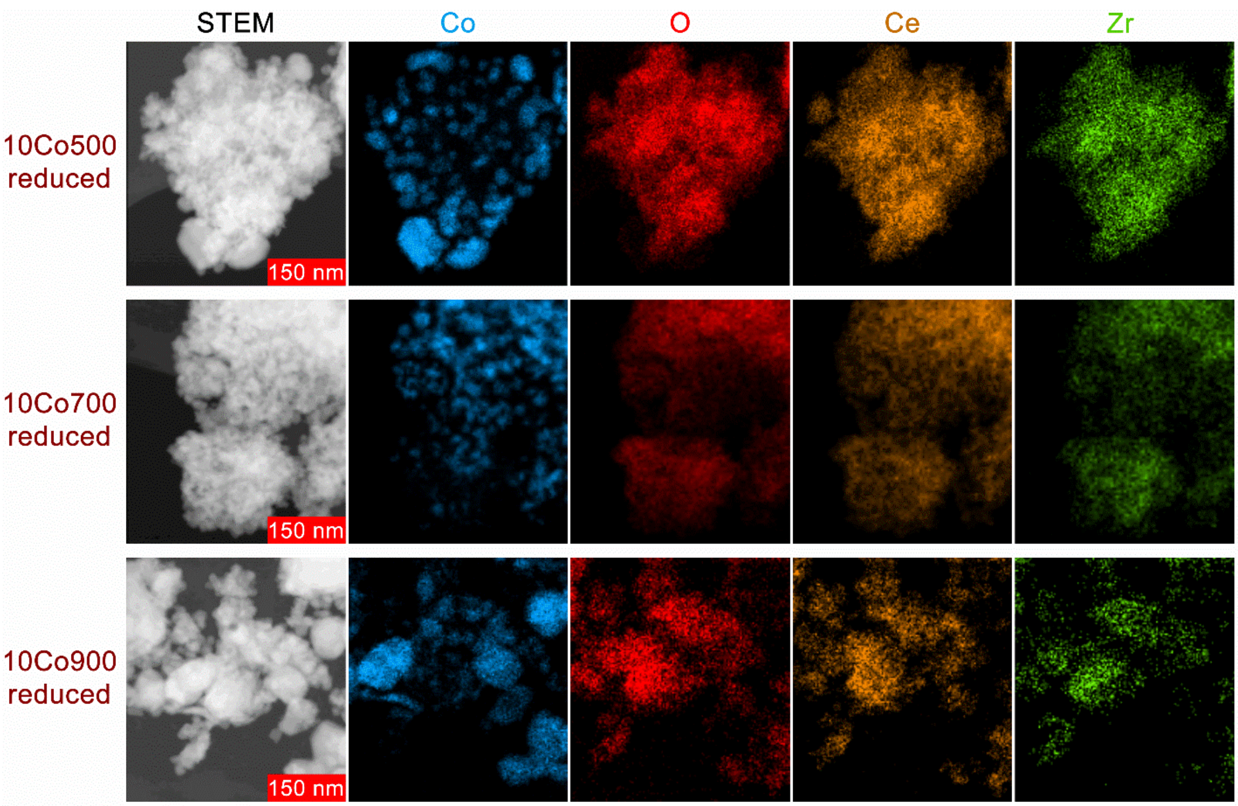
17	20%Ni/Al_2O_3	20	1 bar	350	5.43E-03	[16]
18	3 % Ni/TiO_2	5	1 bar	200 225 250 300	4.10E-04 8.00E-04 1.56E-03 5.31E-03	[17]
19	1.8 % Ni/$CeZrO_4$ 1.9 % Ni@$CeZrO_4$ 2.4 % Ni@SiO_2	5	1 bar	250	2.35E-03 6.15E-03 0.64E-03	[18]
20	5 % Ni/TiO_2	19	1 bar	360	2.43E-02	[19]



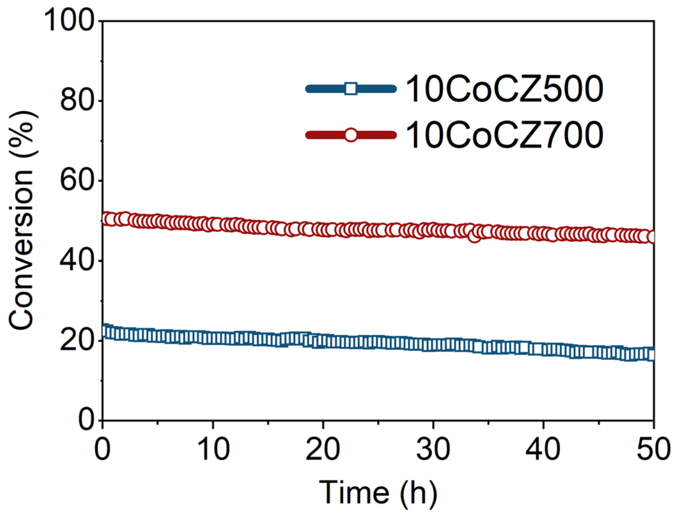
Supplementary Figure 13. XANES spectra obtained during reduction of CoCZ500 (a) and CoCZ700 (b).



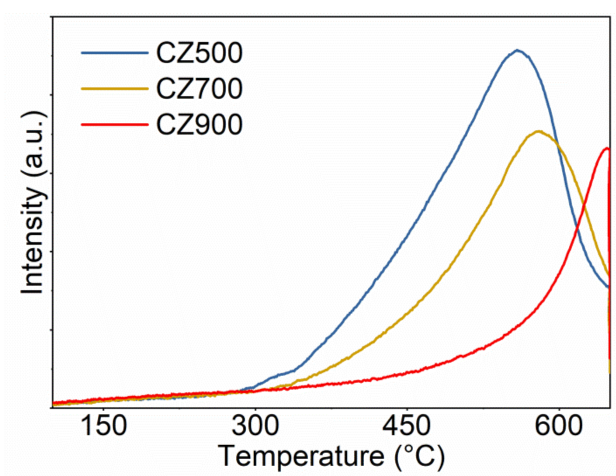
Supplementary Figure 14. Results of linear combination fitting of XANES spectra performed for CoCZ500 and CoCZ700. a) Co_3O_4; b) CoO; c) Co^0.



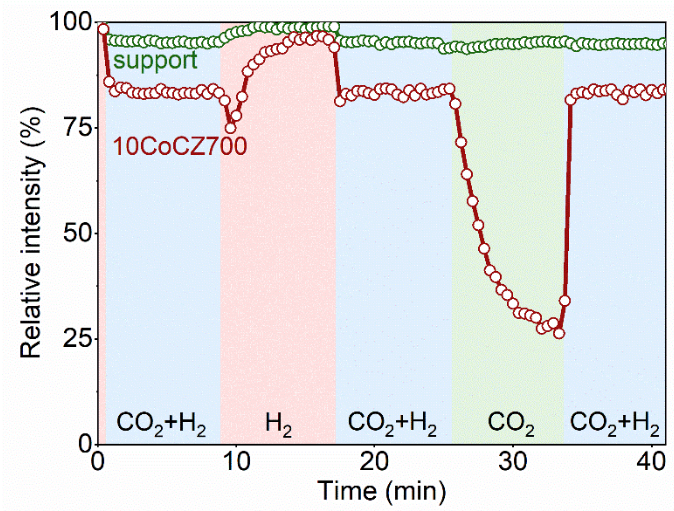
Supplementary Figure 15. STEM images and EDX mapping (Co, O, Ce, Zr) of reduced catalysts: CoCZ500, CoCZ700 and CoCZ900.



Supplementary Figure 16. Stability of CoCZ catalysts in CO_2 hydrogenation. Conditions: 1 bar, 300 °C, 50 mg of catalyst, 5 % CO_2, 20 % H_2, He balance, total flow of 50 mL/min.

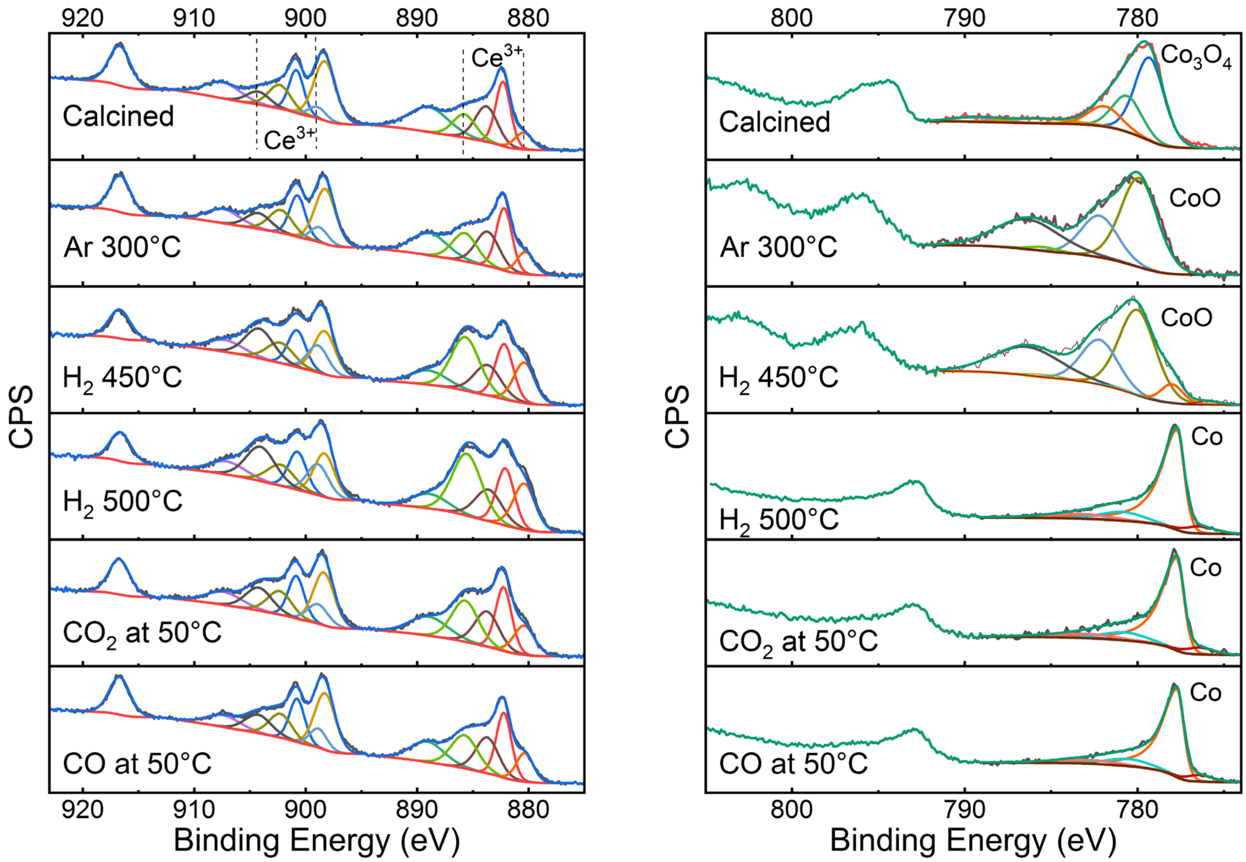


Supplementary Figure 17. TPR profiles of bare ceria-zirconia calcined at 500, 700 and 900 °C.

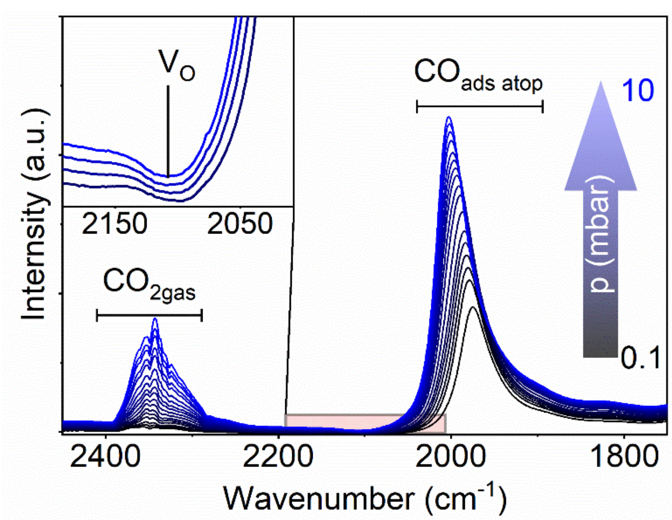


Supplementary Figure 18. Quantification of Ce^{3+} electronic transition band followed by FTIR during transient experiments at 300 °C. Conditions: reduction time 4 h, 1 bar, 300 °C, 12 mg of catalyst (pellet), 2.5 % CO_2, 10 % H_2, He balance, total flow of 200 mL/min.

In order to compare the formation and evolution of oxygen vacancies for bare CZ700 support and CoCZ700 catalyst under reaction conditions (300 °C) we performed transient in situ IR experiments. In these tests the feed composition in the in situ cell was swiftly switched from H_2 to $CO_2 + H_2$, then to H_2, then to $CO_2 + H_2$, then to CO_2 and then finally to $CO_2 + H_2$ again (see Supplementary Figure 18). During all these switches we followed the extent of ceria reduction by quantifying the area of the band at 2115 cm^{-1}. Bare support, reduced at 500 °C, demonstrates negligible changes in Ce^{3+} concentration. Re-oxidation of surface Ce^{3+} species after switching to both the reaction $CO_2 + H_2$ mixture and pure CO_2 amounts to ca. 5 %. In turn, CoCZ700 catalyst with enhanced MSI demonstrated fast dynamic changes during the transient experiments and significant re-oxidation of the ceria support in CO_2 atmosphere and reduction in H_2 atmosphere.

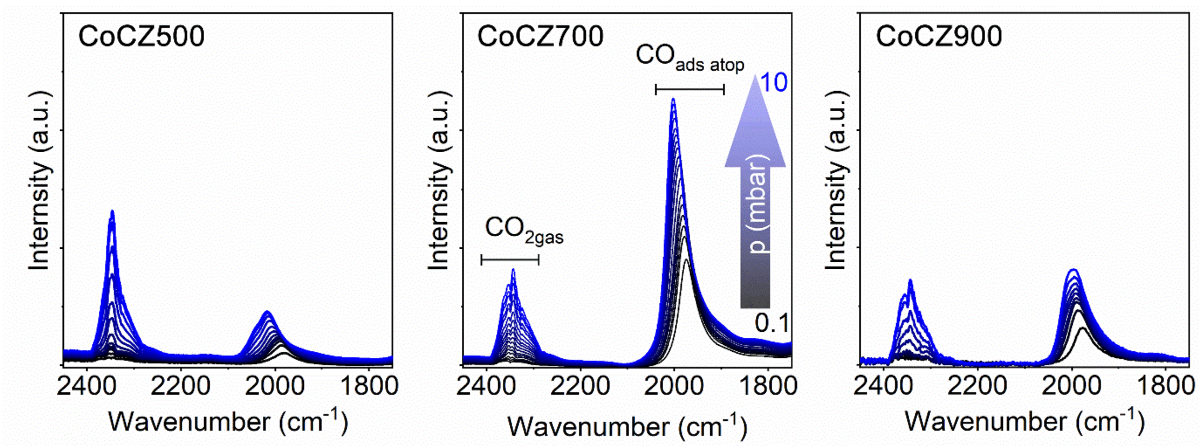


Supplementary Figure 19. NAP-XPS spectra of CoCZ700.



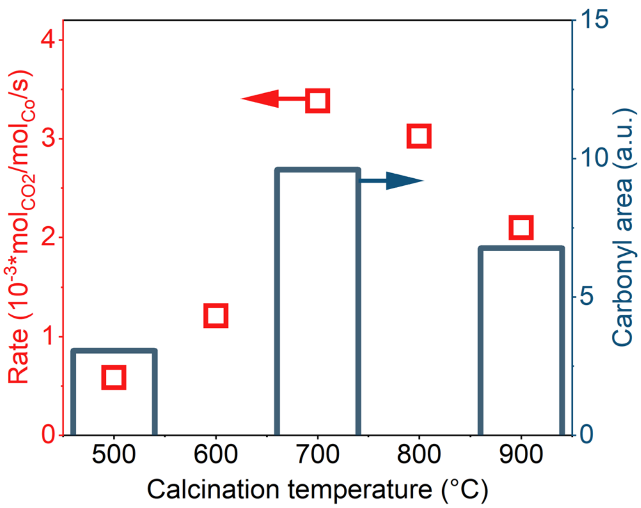
Supplementary Figure 20. Difference IR spectra obtained by exposing the 10Co700 to increasing partial pressure of CO_2 at 50 °C.

Partial re-oxidation of the surface Ce^{3+} is mainly caused by the oxygen spillover from metallic cobalt/cobalt-ceria interface to the ceria surface. In order to obtain a direct evidence of CO_2 dissociation on metallic cobalt sites/cobalt-ceria interface, we exposed CoCZ700 to a different partial pressure of CO_2 (0.1 – 10 mbar) at 50 °C. The presence of intense band of carbonyl species on metallic cobalt surface confirms that CO_2 dissociation takes place and explains the partial oxidation of surface Ce^{3+} observed by NAP-XPS analysis. Moreover, the inset on the Supplementary Figure 20 demonstrates a negative band at 2114 cm^{-1}, pointing at partial re-oxidation of oxygen vacancies upon CO_2 exposure.



Supplementary Figure 21. CO_2 adsorption at 50 °C on reduced catalysts samples followed by IR spectroscopy. Conditions: 50 °C, 12 mg of catalyst, 0.1 – 10 mbar partial pressure of CO_2.

For these experiments we reduced the CoCZ500, CoCZ700 and CoCZ900 catalysts in an *in situ* IR cell (transmission mode) by hydrogen at 500 °C. Then we evacuated the cell and exposed the catalysts to a CO_2 atmosphere of increasing pressure at 50 °C. The results (Supplementary Figure 21) show two groups of vibrational bands appearing after CO_2 exposure. Bands centred around 2350 cm^{-1} correspond to physisorbed (sharp contribution more pronounced for CoCZ500 with higher surface area)[20] and gas phase CO_2. Bands related to carbonyl groups on Co appear around 2000 cm^{-1}. The formation of carbonyls is a result of vacancy-assisted dissociation of CO_2 or dissociation on the metallic Co surface.[21] In this process CO_2 dissociates, yielding a CO molecule adsorbed on Co, while the O atom fills an oxygen vacancy in the ceria support, resulting in the re-oxidation of Ce^{3+} to Ce^{4+} (also observed by NAP-XPS Figs. 4g and 4h in the main text).



Supplementary Figure 22. Correlation between the conversion and ability to dissociate CO_2 for CoCZ catalysts. Values of carbonyl areas are obtained by integration of the corresponding band observed during the exposure of CoCZ catalysts to 10 mbar of CO_2 at 50 °C.

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